
MNL-66(20) AND ACI 318-19 - PLACEMENT RULES WITH

Detailing Rules From The Book

Every reinforcement placement rule the app follows, quoted from the ACI Detailing Manual MNL-66(20) - with printed page, PDF page and figure number, the book's own drawing reproduced, and a note of where WebSketch 3D implements each rule.

Beam, column, wall, slab, foundations, slab on ground.

Source: ACI Detailing Manual MNL-66(20), 502

pages - Section 1 (ACI 315R-18) + Section 2 details

Table of Contents

How to Read This Reference	2
1. Beam	2
2. Column	6
3. Wall	9
4. Slab	12
5. Foundations	13
6. Slab on Ground	16

How to Read This Reference

Each rule below is taken from the ACI Detailing Manual MNL-66(20) itself - the commentary text (Section 1, ACI 315R-18) or the CAD notes of the Section 2 details. Every rule carries its **printed page** and **PDF page** in MNL-66.pdf, the figure number, and the ACI 318-19 clause the book cites. The scanned book page is reproduced under each rule; the grey "In the app" line states where WebSketch 3D implements it. Printed page = PDF page - 9 throughout.

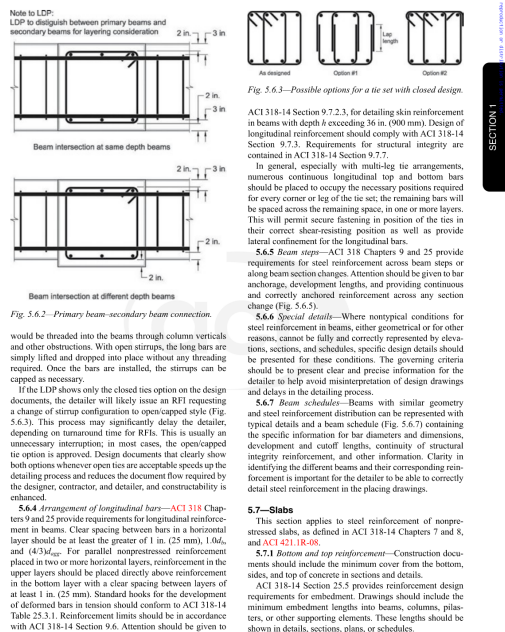
Use it side by side with the app: find the rule, look at the book drawing, then check the same condition in the loaded demo scene.

1. Beam

1.1 Skin reinforcement in deep beams

"ACI 318-14 Section 9.7.2.3, for detailing skin reinforcement in beams with depth h exceeding 36 in. (900 mm)."

Source: MNL-66(20) p. 45 (PDF p. 54), Section 5.6.4; Fig. 5.6.3 on the same page.



Book p. 45 - beam tie options; skin reinforcement per 9.7.2.3.

In the app: Auto when effective depth > 0.9 m: side bars over $h/2$ at max 0.25 m, both faces - buildBeamRebar() skin block.

1.2 Longitudinal bars: development and structural integrity

"Design of longitudinal reinforcement should comply with ACI 318-14 Section 9.7.3. Requirements for structural integrity are contained in ACI 318-14 Section 9.7.7."

Source: MNL-66(20) p. 45 (PDF p. 54), Section 5.6.4.

In the app: Integrity steel 9.8.1.2: min 2+2 continuous bars, hooks at free ends; development by the BM-202 far-side hooks - buildBeamRebar().

1.3 Continuous bars at every tie corner (multi-leg ties)

"In general, especially with multi-leg tie arrangements, numerous continuous longitudinal top and bottom bars should be placed to occupy the necessary positions required for every corner or leg of the tie set; the remaining bars will be spaced across the remaining space, in one or more layers."

Source: MNL-66(20) p. 45 (PDF p. 54), Section 5.6.4.

In the app: The 300 mm leg rule generates inner hoops; continuous bars fill the corners, extras space across in one or two layers - buildBeamRebar() layerPlan().

1.4 Required integrity reinforcement (BM-2 checklist)

"a) At least two of the top and bottom longitudinal bars are continuous... b) Longitudinal integrity reinforcement is enclosed by closed stirrups per Section 25.7.1.6 ... Longitudinal structural integrity bars pass through column bars (9.7.7.3); anchor integrity reinforcement at support to develop f_y (9.7.7.4); splices of positive moment bars must be located near support (9.7.7.5a); splices of negative moment bars must be located near midspan (9.7.7.5b)."

Source: MNL-66(20) p. 91 (PDF p. 100), Figure BM-2 checklist items 13-19.

FIGURE BM-2.1, 2.2, AND 2.3: REQUIRED INTEGRITY REINFORCEMENT FOR PERIMETER BEAMS	
The design professional shall review the following checklist and incorporate project-specific requirements into the details	
INTENDED USE	Typical beam along the perimeter of a building
GENERAL CONSIDERATIONS – NOTE: ALL SECTIONS REFER TO ACI 318-19	
1.	Work Details BM-2.1 through BM-2.3 together
2.	Beam is continuously laterally supported; Section 9.2.3.1
3.	Beam schedule includes:
a.	Beam mark (BM); each beam type is assigned with BM; coordinate BMs on floor plan drawing
b.	Beam dimensions
c.	Top and bottom bar sizes and types
d.	Development length of top bars 1.4 and 1.5
e.	Side bars if beams depth exceeds 36 in. or beam is subjected to torsion
f.	Size, spacing, and type of lateral reinforcement
4.	Minimum flexural reinforcement area in nonprestressed beam satisfies Section 9.6.1
5.	Longitudinal bar spacing satisfies the minimum spacing per Sections 9.7.2.1 and 25.2
6.	Bars closest to the tension face satisfy Sections 9.7.2.2 and 24.3
7.	For beams with $h > 36$ in., side bars, skin reinforcement is provided; Section 9.7.2.3
8.	Flexural bars anchorage, development, extension, embedment, and termination requirements per Sections 9.7.3.2 through 9.7.3.7
9.	At simple supports, at least $1/3$ of $A_{s,dev}$, but not less than two bars satisfy Section 9.7.3.8.1
10.	At continuous supports at least $1/4$ of $A_{s,dev}$, but not less than two bars satisfy Section 9.7.3.8.2
11.	At support at least $1/3$ of A_s , but not less than two bars satisfy Section 9.7.3.8.4
12.	Top bars extend from face of supports a minimum development length; Section 9.7.1.2
13.	Compression bars are laterally supported satisfying Section 9.7.6.4
14.	Integrity reinforcement must satisfy the following; Section 9.7.7.1:
a.	At least $1/4$ of maximum A_s , but not less than two bars are continuous
b.	At least $1/6$ of A_s at support, but not less than two bars are continuous
c.	Longitudinal integrity reinforcement is enclosed by closed stirrups per Section 25.7.1.6
15.	Longitudinal structural integrity bars pass through column bars; Section 9.7.7.3
16.	Anchor integrity reinforcement at support to develop f_y ; Section 9.7.7.4
17.	Splices of positive moment bars must be located near support; Section 9.7.7.5(a)
18.	Splices of negative moment bars must be located near midspan; Section 9.7.7.5(b)
19.	Bars are spliced per Sections 9.7.1.3 and 25.5
20.	Minimum shear reinforcement satisfies Section 9.6.3
21.	Maximum spacing of shear reinforcement along length and across width of beam cross section per Section 9.7.6.2.2
22.	Minimum shear reinforcement bar size satisfies Section 9.7.6.4.2
23.	Concrete reinforcement cover satisfies Sections 9.7.1.1 and 20.5.1
24.	Remove "NOTES TO DESIGN PROFESSIONAL" before placing detail in the Contract Documents
RECOMMENDED REFERENCES	
Concrete International articles: 2, 7, 13, 16, 18, 19, 20, 23	
Refer to Section 3 for articles	
@CivilMethod American Concrete Institute – Copyrighted © Material – www.concrete.org	
aci	

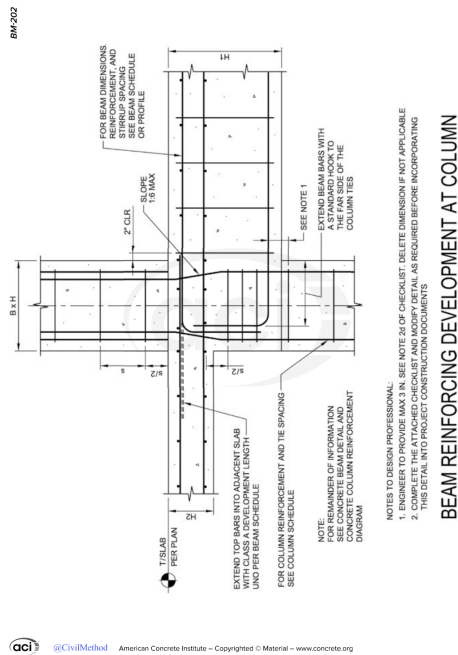
Book p. 91 - BM-2 integrity reinforcement checklist.

In the app: Default-on integrity: closed 135-degree stirrups everywhere; continuous bars pass through the column and anchor with the far-side hook (that IS the develop- f_y anchorage).

1.5 Beam bars at the column - develop to the FAR SIDE (BM-202)

"EXTEND BEAM BARS WITH A STANDARD HOOK TO THE FAR SIDE OF THE COLUMN TIES. EXTEND TOP BARS INTO ADJACENT SLAB WITH CLASS A DEVELOPMENT LENGTH UNO. (notes on the detail; also: 2 in. clear to column bars, 1:6 max offset slope, 2 extra ties at offset)."

Source: MNL-66(20) p. 112 (PDF p. 121), Figure BM-202 "Beam reinforcing development at column".



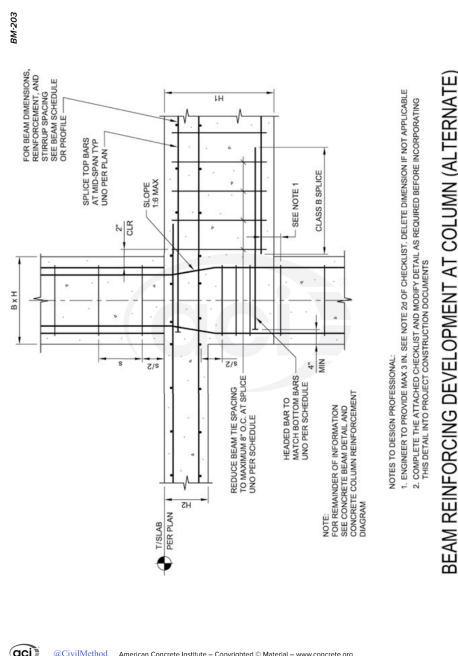
Book p. 112 - BM-202: bars hook at the far side of the column ties.

In the app: Both rows run to center + plan reach - 40 mm cover - tie dia and take the 90-degree hook there; the 8 in connection-zone tie cap covers the splice zone - buildBeamRebar() endSupport().

1.6 Splice development at column, alternate (BM-203)

"SPlice TOP BARS AT MID-SPAN TYP UNO PER PLAN ... REDUCE BEAM TIE SPACING TO MAXIMUM 8 IN. O.C. AT SPLICE UNO PER SCHEDULE ... (headed bar to match bottom bars)."

Source: MNL-66(20) p. 114 (PDF p. 123), Figure BM-203 (alternate development).



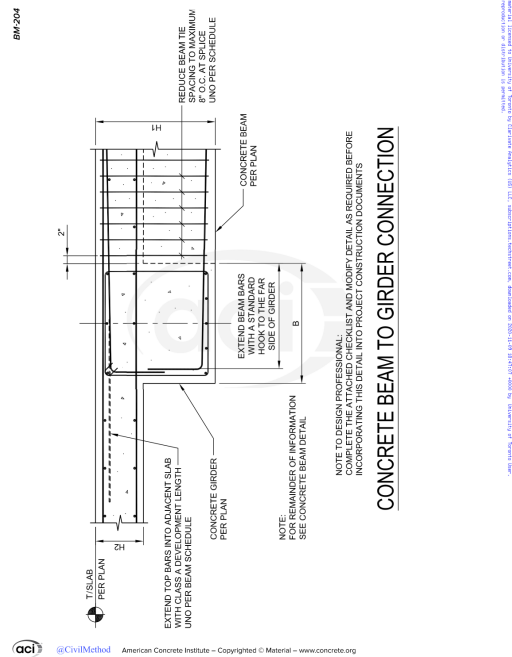
Book p. 114 - BM-203: 8 in max tie spacing at the splice.

In the app: The 8 in (203 mm) max tie spacing is applied through each 0.6 m supported connection zone - buildBeamRebar() BM-203/204 cap.

1.7 Beam-to-girder connection (BM-204/205)

"EXTEND BEAM BARS WITH A STANDARD HOOK TO THE FAR SIDE OF GIRDER. EXTEND TOP BARS INTO ADJACENT SLAB WITH CLASS A DEVELOPMENT LENGTH."

Source: MNL-66(20) p. 116 (PDF p. 125), Figures BM-204/205.



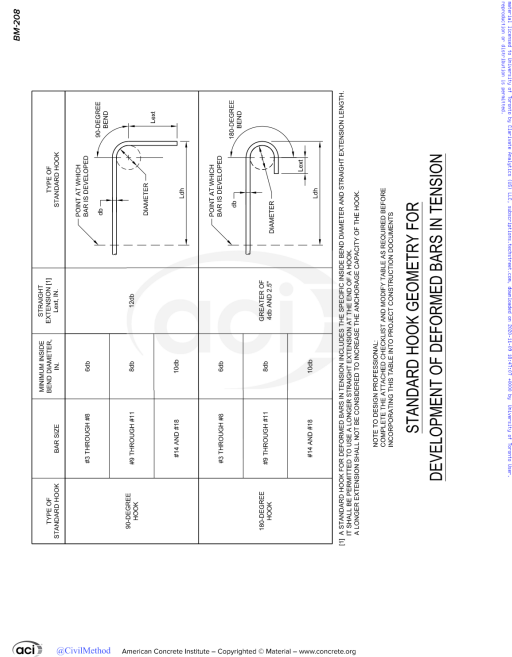
Book p. 116 - BM-204: beam framing into a girder.

In the app: A perpendicular beam crossing at the beam end is detected as the girder: bars hook at its far side (webWidth/2 less covers) - endSupport() girder branch.

1.8 Standard hook geometry (BM-208 table)

"Standard hooks and the 12db extension for 90-degree hooks, by bar size (the BM-208 table gives bend diameters 6/8/10 db by size group)."

Source: MNL-66(20) p. 124 (PDF p. 133), Figure BM-208; ACI 318-19 Table 25.3.1.



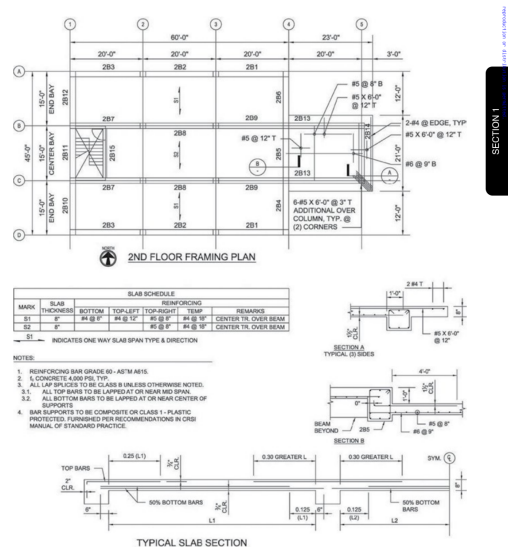
Book p. 124 - BM-208: standard hook table.

In the app: Hook extension 12db at 90 degrees, bend centerline radius 3.5db (6db inside diameter) - addBar() in buildBeamRebar().

1.9 Slab-to-beam bar cutoffs around beams (related rule)

"50 percent of the bottom bars are terminated 0.125 of the clear span distance from the intermediate beam face, except in edge bays where 100 percent of the reinforcement should extend... Top layer flexural reinforcement is placed directly on the beam top layer reinforcement and is required to end with a standard hook at discontinuous ends."

Source: MNL-66(20) p. 47 (PDF p. 56), Fig. 5.7.2 (one-way slab cutoffs).



50 percent of the bottom bars are terminated 0.125 of the clear span distance from the intermediate beam face, except in edge bays where 100 percent of the reinforcement should be on the edge support.

Top layer flexural reinforcement is placed directly on the beam top layer reinforcement and is required to end with a standard hook at discontinuous ends. Support bar details with bar size and spacing or use of temperature bars should be included on the drawings.

@CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

oci

Book p. 47 - Fig. 5.7.2: top bar cutoff details (0.125 Ln terminations).

In the app: Beam extras use $L_n/4$ (top, at supports) and $L_n/8$ (bottom) curtailment; the slabs own their $0.125L_n$ bottom-bar rule - buildSlabRebar(); buildBeamRebar() tCut/bCut.

2. Column

2.1 Tie arrangement schemes

"Column tie arrangement schemes per the figure; ties should extend as close to the compression and tension surfaces as cover requirements permit."

Source: MNL-66(20) p. 43 (PDF p. 52), Fig. 5.5.1; Section 5.5.

TYPE	4 BAR COL	6 BAR COL	8 BAR COL	10 BAR COL	12 BAR COL
A					
B					
C					
D					

Fig. 5.5.1—Column tie arrangement schedule.

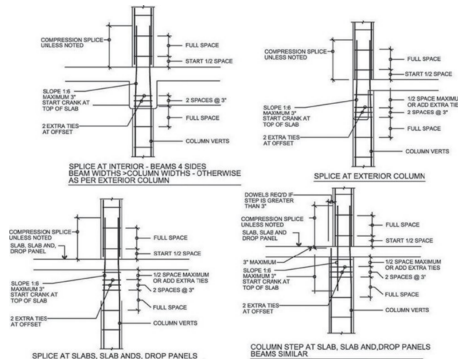


Fig. 5.5.3—Column details at steps and transitions.

(V). If there is a column offset greater than 3 in. (75 mm), a slope transition is not allowed and is to be made with separate dowels adjacent to the offset column faces. Where vertical bars are offset, horizontal support should be provided throughout by ties, spirals, or parts of the floor construction. If transverse reinforcement is provided, they should be

@CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

oci

Book p. 43 - Fig. 5.5.1: column tie arrangement schemes.

In the app: Closed tie set from stirrupPath() at the covers; every corner bar engaged by a tie bend - buildColumnCage() ties.

2.2 Tie types schedule - staggered hooks

"COLUMN TIE TYPE SCHEDULE: 4/6/8/10/12-bar columns; HOOKS STAGGERED AROUND PERIMETER."

Source: MNL-66(20) p. 132 (PDF p. 141), Figure COL-20.

COL-20	12 BAR COL				
	10 BAR COL				
	8 BAR COL				
	6 BAR COL				
	4 BAR COL				
	A	B	C	D	

NOTES TO DESIGN PROFESSIONAL:
- COMPLETE THE ATTACHED CHECKLIST AND MODIFY SCHEDULE AS REQUIRED BEFORE INCORPORATING THIS SCHEDULE INTO PROJECT CONSTRUCTION DOCUMENTS

COLUMN TIE TYPE SCHEDULE

HOOKS STAGGERED AROUND PERIMETER

@CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

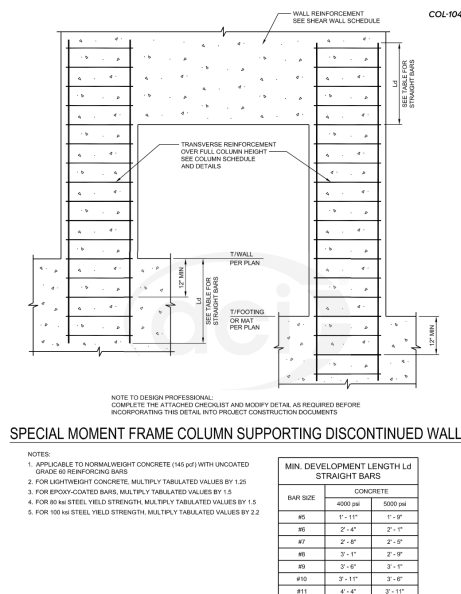
Book p. 132 - COL-20: column tie type schedule.

In the app: 135-degree lap hooks alternate around the section between consecutive ties (stirrupPath alternation + seismic lap alternation).

2.3 Development length table (COL-104)

"MINIMUM DEVELOPMENT LENGTH L_d STRAIGHT BARS - table by bar size for 4000/5000 psi (e.g. 1 ft-9 in. / 1 ft-11 in. for small bars)."

Source: MNL-66(20) p. 140 (PDF p. 149), Figure COL-104.



ACI @CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

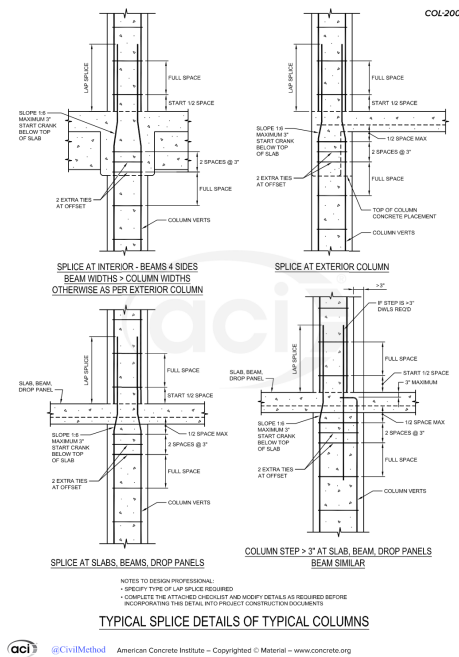
Book p. 140 - COL-104: development length table.

In the app: Column lap = $1.3 \times 47.5 \times \text{psi-s} \times \text{db}$ (Class B from L_d , Grade 60, 4 ksi; 0.69 m for 14 mm, 0.79 m for 16 mm) - verticalBar() splice block.

2.4 Splice locations (COL-200)

"SPlice AT SLABS, BEAMS, DROP PANELS ... splice at interior/exterior column ... COLUMN STEP GREATER THAN 3 IN. AT SLAB: slope 1:6 maximum, start crank below top of slab, 2 EXTRA TIES AT OFFSET ... CLASS B LAP SPLICE."

Source: MNL-66(20) p. 142 (PDF p. 151), Figure COL-200 (with COL-101 notes).



Book p. 142 - COL-200: splice at slabs/beams.

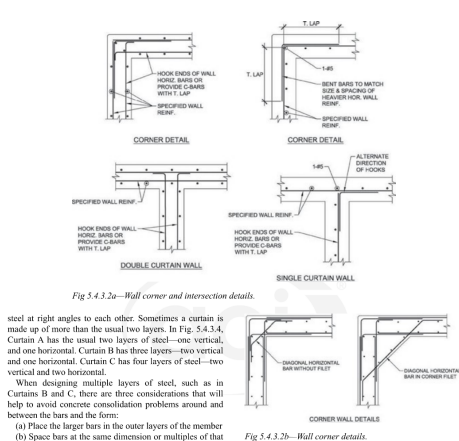
In the app: Lap mode: two pieces just above the floor, Class B lap, alternate bars staggered half a lap, 2 extra ties through the splice zone - verticalBar() + the splice-tie pass.

3. Wall

3.1 Corners and intersections

"Wall corner and intersection reinforcement - single/double layer options with tension lap splices and corner bars."

Source: MNL-66(20) p. 40 (PDF p. 49), Fig. 5.4.3.2a.



steel at right angles to each other. Sometimes a curtain is made up of more than the usual two layers. In Fig. 5.4.3.4, Curtain A has the usual two layers of steel—one vertical, and one horizontal. Curtain B has three layers—two vertical and one horizontal. Curtain C has four layers of steel—two vertical and two horizontal.

When designing multiple layers of steel, such as in Curtains B and C, there are three considerations that will help to avoid concrete consolidation problems around and between the bars and the form:

- Place the larger bars in the outer layers of the member
- Space bars at the same dimension or multiples of that dimension in all layers
- Use only one bar size in a given layer

Frequently, designing curtains with more than the usual two layers of steel will cause spatial problems that affect clearances and constructability. Attention should be given to potential problems that can occur at wall corners and when staggering laps. The designer should consider using larger bar sizes, higher-strength steel, closer spacing, bundled bars, or a combination of these to avoid the need for multiple layers of reinforcing bars. Space between parallel bars should be adequate for the maximum size aggregate used and a vibrator to be inserted between the bars for consolidating the concrete.

5.4.3.5 Design details at joints and waterstops—The design details for reinforcement continuity or termination at wall joints with or without waterstops should be provided. Construction joints are vertical or horizontal joints that are intentionally created between two successive placements of concrete. Shear keys can be used to increase the shear resistance at the joint. If keys are not used, the surface of the first pour should be cleaned and roughened as required per the construction documents before the next concrete pour. Keys are commonly formed in the wall base to give the stem more sliding resistance. Wall reinforcement is placed continuously across the joint (Fig. 5.4.3.5a).

Contraction joints are joints formed, sawed, or tool-grooved in a concrete structure to create a weakened plane to regulate the location of cracking resulting from the dimensional change of different parts of the structure. In walls, vertical contraction joints are usually about 14 in. (5 to 6 mm) wide and approximately 1/2 to 3/4 in. (12 to 20 mm) deep, and are provided at various intervals, depending on wall height, thickness, and amount of reinforcement, but usually not exceeding approximately 30 ft (9 to 10 m). Wall reinforcement is placed continuously across the joint (Fig. 5.4.3.5b).

Book p. 40 - Fig. 5.4.3.2a: wall corner and intersection details.

In the app: Wall mesh is continuous through the run; intersections of hosted walls carry the dowel/trim steel of chapters 3.3/3.4.

3.2 Reinforcement curtains

"Wall reinforcement curtains (two layers): vertical and horizontal bars each face."

Source: MNL-66(20) p. 41 (PDF p. 50), Fig. 5.4.3.4.

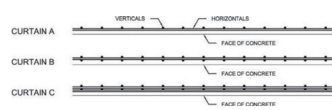


Fig. 5.4.3.4—Wall reinforcement curtain arrangements.

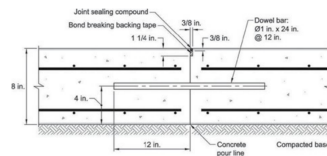


Fig. 5.4.3.5a—Detail of a wall construction joint.

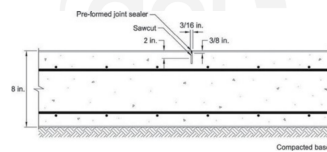


Fig. 5.4.3.5b—Detail of a wall contraction joint.

Expansion joints are a separation provided between adjacent sections of a concrete structure to allow movement due to dimensional increases and reductions of the adjacent sections, and through which some or all the bonded reinforcement is interrupted. In walls, horizontal grooved dowels are usually placed across vertical joints to tie adjacent sections together. Expansion joints are located at various intervals, depending on wall dimensions, but should not be spaced more than 75 to 90 ft (25 to 30 m) apart. Wall reinforcement is discontinuous across the joint, not placed across the joint but rather terminated at each joint face (Fig. 5.4.3.5c). Waterstops are thin sheets of metal, rubber, plastic, or other material inserted across a joint to obstruct the seepage of water through the joint. They should be embedded equally in each side of the joint. There are various sections that can be used for the different types of joints named previously. In addition to molded waterstops that are embedded in concrete across the joint, there are adhesive-type waterstops that do not interfere with the reinforcement. These are strips of rubber and mastic that adhere to the surface of set concrete. The next placement of concrete compresses the rubber strip to form a watertight seal between the two placements of concrete. Steel reinforcement design details should be developed with consideration of the waterstop to be used in the joint. An example design detail for a waterstop and the effect of depressing the slab top reinforcement is shown in Fig. 5.4.3.5d.

@CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

aci

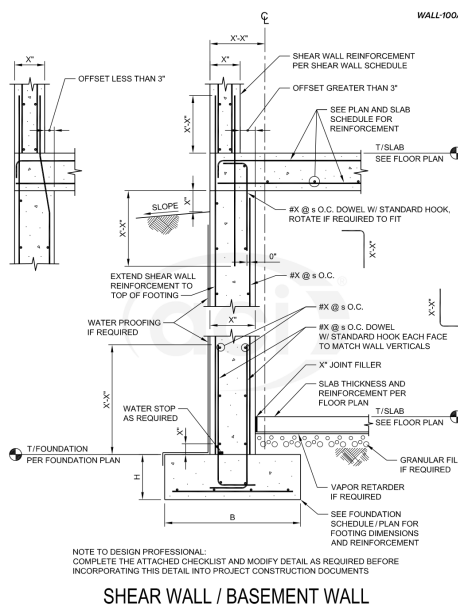
Book p. 41 - Fig. 5.4.3.4: wall reinforcement curtains.

In the app: twoCurtains option - each curtain at the cover on its face - buildWallRebar().

3.3 Dowels from the support below (WALL-100A)

"#X @ s O.C. DOWEL WITH STANDARD HOOK, ROTATE IF REQUIRED TO FIT ... (dowels matching the wall verticals, cast with the slab/footing below)."

Source: MNL-66(20) p. 154 (PDF p. 163), Figure WALL-100A.



SHEAR WALL / BASEMENT WALL

@CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

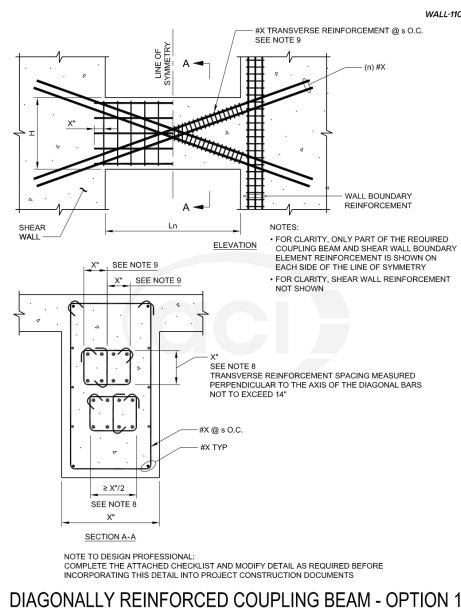
Book p. 154 - WALL-100A: dowels with standard hooks.

In the app: One L-dowel per vertical station per curtain where a slab/footing top meets the wall base; leg inside the host, Class B lap into the wall - buildWallRebar() dowelHost block.

3.4 Boundary elements (WALL-110)

"WALL BOUNDARY REINFORCEMENT ... transverse reinforcement spacing not to exceed 14 in. measured perpendicular to the diagonal bars; diagonally reinforced coupling beam."

Source: MNL-66(20) p. 168 (PDF p. 177), Figure WALL-110.



ACI @CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

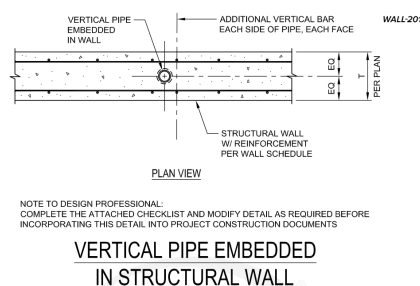
Book p. 168 - WALL-110: boundary element reinforcement.

In the app: boundary option: 2 concentrated verticals inside each wall end per curtain - buildWallRebar() boundary block.

3.5 Circular pipe opening (WALL-201)

"VERTICAL PIPE EMBEDDED IN STRUCTURAL WALL ... ADDITIONAL VERTICAL BAR EACH SIDE OF PIPE, EACH FACE."

Source: MNL-66(20) p. 184 (PDF p. 193), Figure WALL-201.



ACI @CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

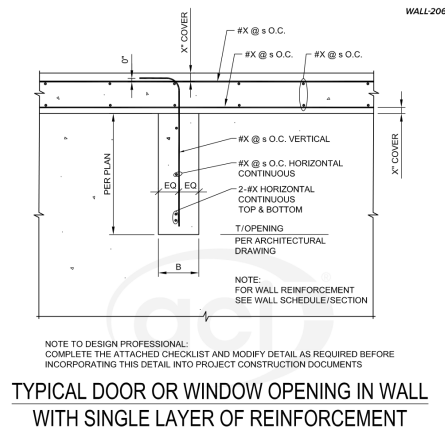
Book p. 184 - WALL-201: pipe penetration.

In the app: shape:"circle" openings: mesh splits around the disc + trim pairs each side (the additional bars) - buildWallRebar() circular branch.

3.6 Openings in walls (WALL-206/207/208)

"TYPICAL DOOR OR WINDOW OPENING IN WALL: 2-#X HORIZONTAL CONTINUOUS TOP AND BOTTOM ... (jambs get #X @ s O.C. vertical; WALL-208 adds one #X x 48 in. diagonal bar per corner per layer, 2 ft-0 in. minimum extension, hook bars when 24 in. length is not available)."

Source: MNL-66(20) p. 194 (PDF p. 203), Figure WALL-206; WALL-208 at p. 201 (PDF p. 210).



In the app: Trim pairs at head/sill/jambs (24 in development), 48 in diagonals at each corner per curtain, clamped to fit - buildWallRebar() openings loop.

4. Slab

4.1 Corner reinforcement (SLAB-200/201)

"Slab corner reinforcement options - top and bottom bars both directions at exterior corners, extending the longer span/5 from the corner."

Source: MNL-66(20) p. 64 (PDF p. 73), Figures SLAB-200/201; ACI 318-19 8.6.1.2.

5.1 Staggered lap splices (general)

"Staggered lap splices and staggered bar terminations - the general placement rule for all elements."

Source: MNL-66(20) p. 36 (PDF p. 45), Fig. 5.2.1.

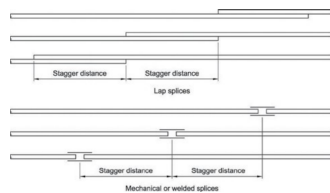


Fig. 5.2.1—Staggered lap splices and staggered mechanical splices.

be no lap. The design drawings should clearly indicate if a lap is intended and if so, dimension the lap (Fig. 5.2.3). Where standard hooks would be too long to fit within the concrete member, the desired hook should be clearly dimensioned on the drawing.

5.2.4 Skewed bars—In plan views of trapezoidal-shaped slabs, the LDP should clearly indicate if the main bars are perpendicular to the parallel sides or parallel to the sloped sides (Fig. 5.2.4). Also indicate if the spacing of the bars is measured at right angles to the main bars or along the skew. These points should also be made clear for triangular or other irregular-shaped slabs.

Top steel over beams in skewed slabs can conflict with top bars from adjacent regular-shaped slabs and could cause layering and clearance problems. These conditions should be considered and addressed by the LDP.

5.2.5 Termination of vertical bars—The design drawings should always indicate how vertical bars are to be terminated. If the bars are to be hooked, there are several considerations as follows:

- If the hook is not a standard hook, the length should be indicated.
- If the hook is located in either of the top layers or beneath both layers.
- If the member into which the hook is to be embedded is sufficiently deep enough to accommodate the curvature of the bend.
- If the direction of the hook is critical, it should be clearly indicated.

If nothing is indicated, hook direction will not be noted on the shop drawings and will likely be placed based on other field constraints. Member intersections can require additional consideration of bar interferences.

5.2.6 Beam stirrups—On a beam schedule, it should be clear whether a stirrup callout—for example, 3 @ 6 in.; 5 @ 9 in.; rem @ 12 in. (3 @ 150 mm; 5 @ 225 mm; rem @ 300 mm)—is referring to the number of stirrups or the number of spaces. If spacing of the first stirrup from a support is

critical, it should be indicated, otherwise it will be located one-half space from the face of support.

In multiple tie sets, indicate if these are to be a series of nested stirrups or a series of interlocking stirrups. If open stirrups are indicated, if possible, try to provide a continuous top bar in each stirrup hook. Consider a greater quantity of smaller longitudinal bars rather than fewer larger bars.

In very narrow beams, determine if the stirrup hooks will allow the top bars to fit within the tie. If critical, indicate if longitudinal bars are to be lapped in a vertical or horizontal plane. At intersecting beams, indicate if the stirrups run through one beam or the other, or through both beams.

5.3—Foundations

This section applies to nonprestressed steel reinforcement of shallow and deep structural foundations as defined by ACI 318-14 Chapter 13 and includes piles, drilled piers, and caissons as further discussed in ACI 318-14 and ACI 543R. Refer to Section 5.5 of ACI 543R-12 for the supplemental details of deep foundations in structures assigned to Seismic Design Categories D, E, and F as prescribed by ACI 318-14 Section 18.13.4.

5.3.1 Types—Generally, foundation systems are categorized as either shallow foundations or deep foundations. Shallow foundations include such types as strip footings; isolated, spread, or pad footings; mat foundations; and grade beams. Caps are typically shallow elements transferring load to deep foundations.

5.3.2 Bar arrangements—Bar arrangements vary widely based on both the foundation type and the lateral, gravity, and torsional loadings imposed on the foundations. The example reinforcing details described in this section are common arrangements. The LDP should specify appropriate bar arrangements for the various conditions.

Shallow foundations such as strip, isolated, and mat foundations often have a bar arrangement similar to that of a slab with a bottom mat, a top mat, or both bottom and top mat of reinforcement with bars in each direction. In some instances,

@CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

Book p. 36 - Fig. 5.2.1: staggered lap splices.

In the app: Column and wall cages stagger alternate bars (half lap); beam extra bars curtail at different lengths.

5.2 Development and lap splice tables

"Development and lap splice tables by bar size and concrete strength."

Source: MNL-66(20) p. 37 (PDF p. 46), Fig. 5.2.2.

LAP SPlice AND DEVELOPMENT LENGTH (SCHEDULE INCHES)									
F _c = 4,000 PSI									
BAR SIZE	COMP	TENSION			LTE	LTS	LTS OTR	OTR TOP	LTS TOP
	LCE	LCS	LDH	LDH					
#3	8	12	6	15	26	20			
#4	10	15	7	20	34	26			
#5	13	19	9	26	43	33			
#6	16	23	11	30	51	40			
#7	18	27	12	44	75	58			
#8	21	30	14	51	86	66			
#9	23	34	16	57	97	74			
#10	26	38	18	64	109	84			
#11	29	43	20	72	121	93			
LCE—COMPRESSION EMBEDMENT LENGTH LCS—COMPRESSION LAP SPlice LENGTH LDH—HOOK DEVELOPMENT LENGTH LTE—TENSION EMBEDMENT LENGTH LTS—TENSION LAP SPlice LENGTH									

LAP SPlice AND DEVELOPMENT LENGTH (SCHEDULE INCHES)									
F _c = 4,500 PSI									
BAR SIZE	COMP	TENSION			LTE	LTS	LTS OTR	OTR TOP	LTS TOP
	LCE	LCS	LDH	LDH					
#3	8	12	6	14	23	18			
#4	10	15	6	18	30	23			
#5	12	18	8	22	36	26			
#6	15	23	9	27	45	35			
#7	17	27	11	30	56	31			
#8	19	30	13	45	76	38			
#9	22	34	14	51	85	46			
#10	24	38	16	57	96	54			
#11	27	43	18	63	107	62			
LCE—COMPRESSION EMBEDMENT LENGTH LCS—COMPRESSION LAP SPlice LENGTH LDH—HOOK DEVELOPMENT LENGTH LTE—TENSION EMBEDMENT LENGTH LTS—TENSION LAP SPlice LENGTH									

LAP SPlice AND DEVELOPMENT LENGTH (SCHEDULE INCHES)									
F _c = 5,000 PSI									
BAR SIZE	COMP	TENSION			LTE	LTS	LTS OTR	OTR TOP	LTS TOP
	LCE	LCS	LDH	LDH					
#3	8	12	6	13	22	17			
#4	9	15	6	17	29	22			
#5	12	18	7	21	36	26			
#6	14	23	9	25	43	33			
#7	16	27	10	27	53	48			
#8	18	30	12	42	72	55			
#9	21	34	13	48	81	62			
#10	23	38	15	54	91	70			
#11	26	43	17	60	101	79			
LCE—COMPRESSION EMBEDMENT LENGTH LCS—COMPRESSION LAP SPlice LENGTH LDH—HOOK DEVELOPMENT LENGTH LTE—TENSION EMBEDMENT LENGTH LTS—TENSION LAP SPlice LENGTH									

Fig. 5.2.2—Development and lap splice table.

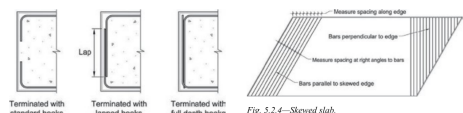


Fig. 5.2.3—Possible hook configurations at edge of raft footing.

@CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

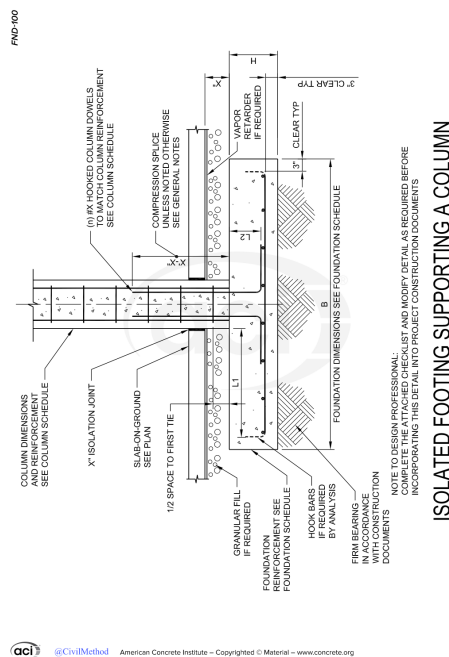
Book p. 37 - Fig. 5.2.2: development and lap splice tables.

In the app: The Class B lap (1.3 x ld) used everywhere: ld = 47.5 x psi-s x db at Grade 60 / 4 ksi.

5.3 Spread footing with dowels (FND-100)

"Footing with column dowels - dowels develop into the column with a standard hook into the footing; minimum cover and mat support."

Source: MNL-66(20) p. 218 (PDF p. 227), Figures FND-100 series.



ACI CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

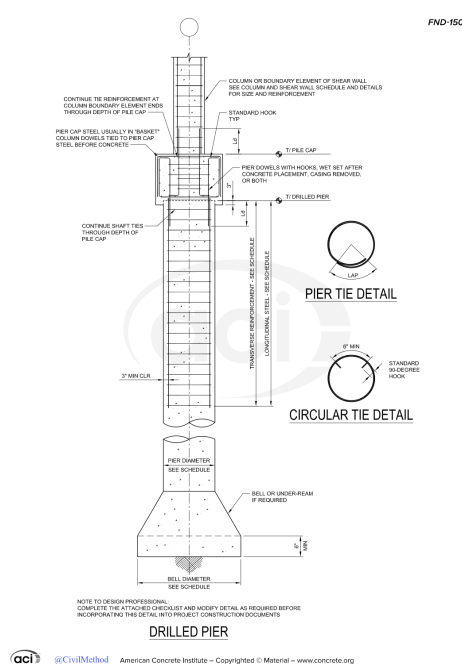
Book p. 218 - FND-100: footing with dowels.

In the app: L-starters with the hook leg in the footing, rising the Class B lap (0.69 m for 14 mm) - buildFootingRebar() starters.

5.4 Drilled pier (FND-150)

"T/O DRILLED PIER ... CONTINUE SHAFT TIES THROUGH DEPTH OF PILE CAP ... STANDARD 90-DEGREE HOOK (column bars onto the shaft steel)."

Source: MNL-66(20) p. 256 (PDF p. 265), Figure FND-150.



ACI CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

Book p. 256 - FND-150: drilled pier.

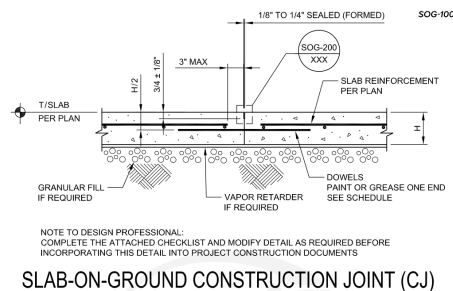
In the app: Circular pads auto-detect: ring verticals at max 200 mm + circular ties through the cap; the shaft steel rises the lap above the cap for the column splice - buildFootingRebar() pier branch.

6. Slab on Ground

6.1 SOG at the wall (SOG-100/102)

"SLAB-ON-GROUND REINFORCEMENT PER PLAN (CONTINUOUS THROUGH THE THICKENING) ... 4-#X CONTINUOUS BOTTOM ADDITIONAL BARS ... DOWELS TO MATCH WALL VERTICAL REINFORCEMENT (ALTERNATE HOOKS)."

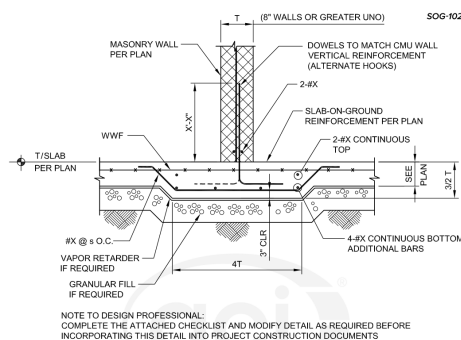
Source: MNL-66(20) p. 298-302 (PDF p. 307/311), Figures SOG-100 and SOG-102.



SLAB-ON-GROUND CONSTRUCTION JOINT (CJ)

ACI @CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

Book p. 298 - SOG-100: slab-on-ground at wall.



INTERIOR THICKENED SLAB-ON-GROUND
FOOTING AT CMU NONBEARING WALL - OPTION 1

ACI @CivilMethod American Concrete Institute - Copyrighted © Material - www.concrete.org

Book p. 302 - SOG-102: thickened edge bars.

In the app: SOG toggle: mesh unbroken through the edge + 2 continuous edge bars per side lapped $\min(0.6 \text{ m}, L/4)$ past corners - buildSlabRebar() sog block.